

RoCEv2 800G Data Center AI Fabric Architecture Specification

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Chapter 1: Architectural Overview

Defines lossless RDMA over Converged Ethernet (RoCEv2) for 800Gb/s spine-leaf topologies powered by Broadcom Tomahawk 5 (SKU-SW-TH5) switching silicon and PAM4 OSFP optical transceivers.

Chapter 2: Mechanical & Thermal Specifications

2.1 Power Envelope: Designed for high-density AI clusters supporting up to 48kW per rack busbar at 48V DC nominal bus architecture.

2.2 Thermal Limits: Maximum inlet water temperature for direct-to-chip liquid cooling manifolds is 32°C. Return water temperature is maintained at 55°C to 65°C, optimized for district heating heat-exchange coupling.

2.3 Form Factor & Dimensions: Compliant with Open Compute Project (OCP) Open Rack v3 (ORV3) 21-inch chassis pitch and 44OU vertical space.

Chapter 3: Interconnect, Network & Memory

3.1 Optical Interconnects: 800Gb/s OSFP optical transceivers utilizing PAM4 modulation over DR8 parallel single-mode fiber.

3.2 Memory Architecture: High Bandwidth Memory (HBM3e) delivering up to 5.3 TB/s memory bandwidth per accelerator node, paired with DDR5-5600 ECC system memory.

Chapter 4: Software Stack & Driver Compatibility

4.1 Runtime Environment: Full open-source driver stack support including AMD ROCm 6.2 and ROCm 6.3 with PyTorch 2.4+ native kernel execution, eliminating proprietary closed-source CUDA runtime dependencies.

4.2 Validated Large Language Models: Validated out-of-the-box for Mistral-Large, Llama-3-70B, and Mixtral 8x22B inference and fine-tuning pipelines under standard ROCm containerized environments.